Probabilistic Regression

Bridging Classification and Regression for Robust Uncertainty Quantification

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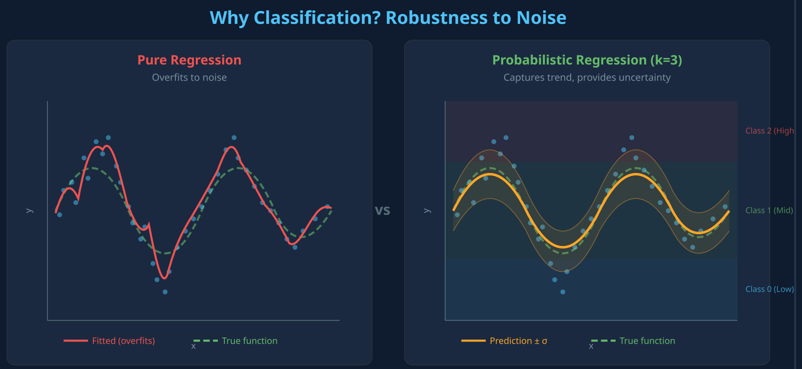
A Gaussian mixture framework that adapts model complexity to signal-to-noise ratio

Motivation & Problem Statement

Standard regression minimises squared error — producing **point estimates only**, with no uncertainty.

$$L = \Sigma\,(y_i - f(x_i))^2 \;\rightarrow\; \text{just } \hat{y}\text{, no } \sigma^2$$

- In noisy regimes (low SNR), the model overfits to noise
- We want **p(y | x)** — a full predictive distribution
- Key insight: *classification is inherently more robust to noise than regression*



Core Idea — Discretisation as Regularisation

Step 1 — Discretise

Partition target space into k classes via quantiles:

$$C_j = \{y : q_{j-1} \le y < q_j\}, \quad j = 1,\ldots,k$$

Step 2 — Classify

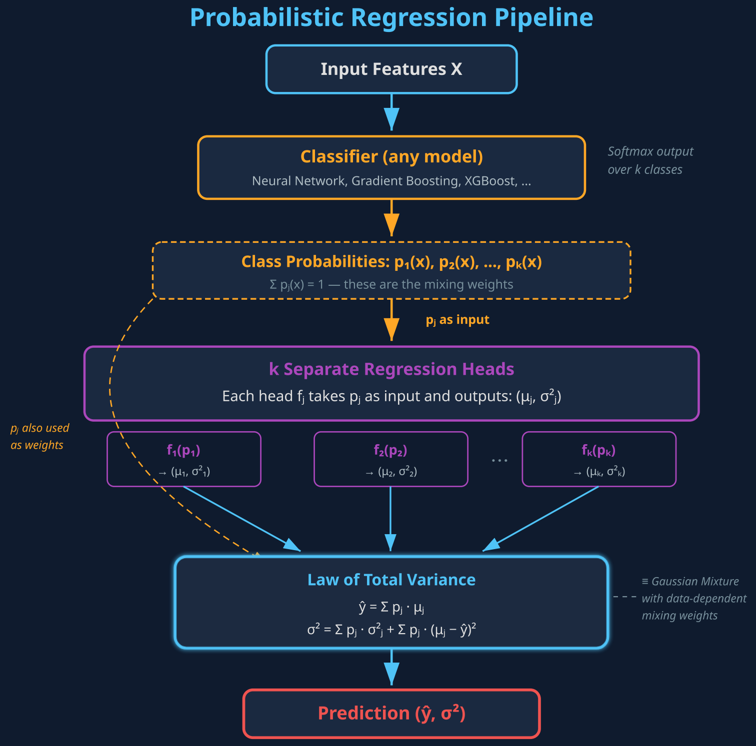
Train a classifier for class probabilities:

$$p_j(x) = P(y \in C_j \mid x), \quad \Sigma\, p_j = 1$$

Step 3 — Regress per class

Each head maps probability to Gaussian params:

$$f_j : p_j \mapsto (\mu_j, \sigma_j^2)$$



Equivalent to a **Gaussian Mixture Model** with data-dependent mixing weights: $p(y|x) = \Sigma_j\, p_j(x) \cdot \mathcal{N}(y; \mu_j(x), \sigma_j^2(x))$

Prediction & Uncertainty via Law of Total Variance

POINT PREDICTION — LAW OF TOTAL EXPECTATION

$$\hat{y}(x) = E[Y \mid X] = \sum_{j=1}^{k} p_j(x) \cdot \mu_j(x)$$

UNCERTAINTY — LAW OF TOTAL VARIANCE

$$\mathrm{Var}(Y \mid X) = \sum_j p_j \cdot \sigma_j^2 + \sum_j p_j \cdot (\mu_j - \hat{y})^2$$

Within-class (aleatoric)

$E[\mathrm{Var}(Y|C)]$ — uncertainty *within* each class

Between-class (epistemic)

$\mathrm{Var}(E[Y|C])$ — how much class means *disagree*

This decomposition is **exact** under the mixture model — not an approximation. Provides calibrated prediction intervals: $\hat{y} \pm z \cdot \sigma$

k as Signal-to-Noise Ratio Proxy

The number of classes **k** is an interpretable hyperparameter controlling the *precision–robustness trade-off*.

Regime	k	Behaviour	Intuition
High noise (low SNR)	k = 2	Binary: above/below median	Easiest classification, most robust
Medium noise	k = 3–5	Moderate resolution	Balanced
Low noise (high SNR)	$k \to \infty$	Approaches pure regression	Each bin ≈ 1 value

Optimal k is inversely related to noise level

As k increases: classification accuracy ↓ (harder task) but regression resolution ↑ (finer bins)

Training Objective & Results

TOTAL LOSS (TRAINED END-TO-END)

$$L = L_{regression} + L_{classification}$$

REGRESSION — GAUSSIAN LIKELIHOOD

$$p(y \mid x) = N(y \,;\, \hat{y}, \sigma^2)$$

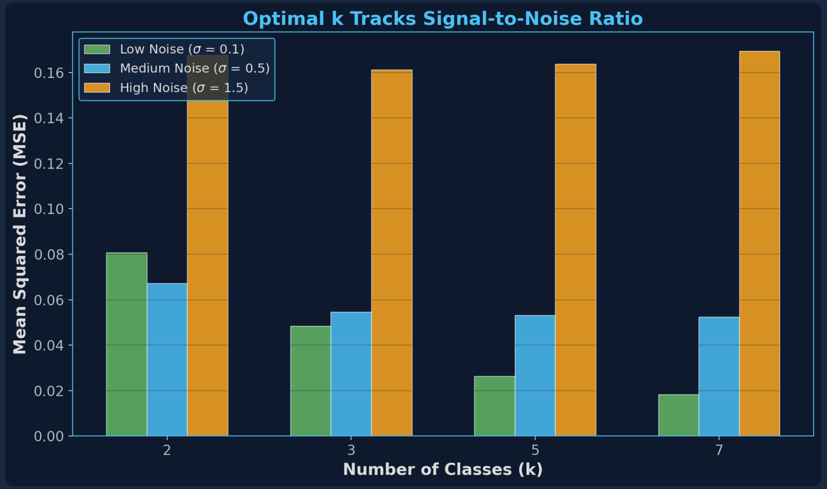
$$L_{reg} = ½ \, [\, \log(2\pi) + \log \sigma^2 + (y - \hat{y})^2 / \sigma^2 \,]$$

CLASSIFICATION — CROSS-ENTROPY

$$L_{cls} = - \sum_j \mathbb{1}[y \in bin_j] \cdot \log p_j(x)$$

L_{cls} → good class assignments | L_{reg} → accurate predictions & calibrated σ^2

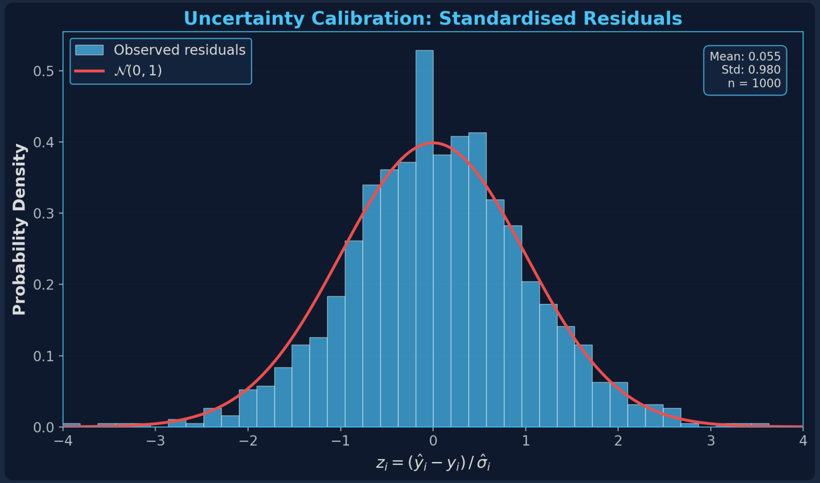
Optimal k tracks SNR



At high noise, small k wins. At low noise, large k wins.

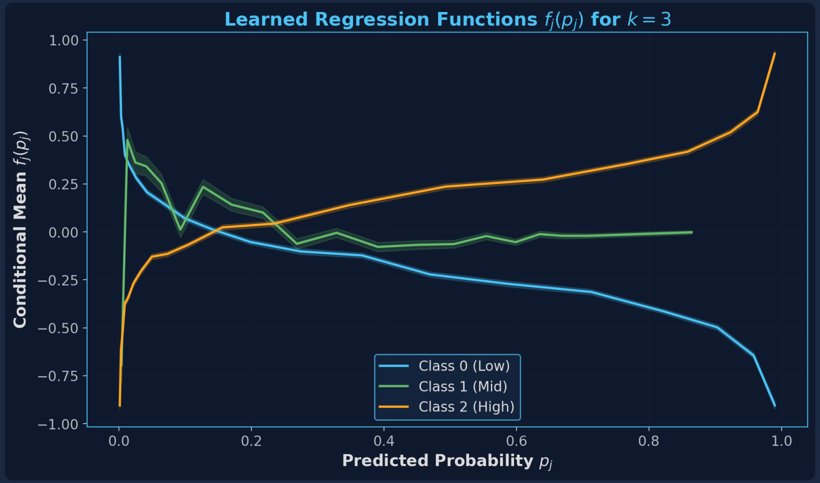
Results — Calibration & Learned Functions

Calibrated Uncertainty



Standardised residuals $z_i = (\hat{y}_i - y_i) / \hat{\sigma}_i$ closely match $\mathcal{N}(0,1)$ — uncertainty is well-calibrated

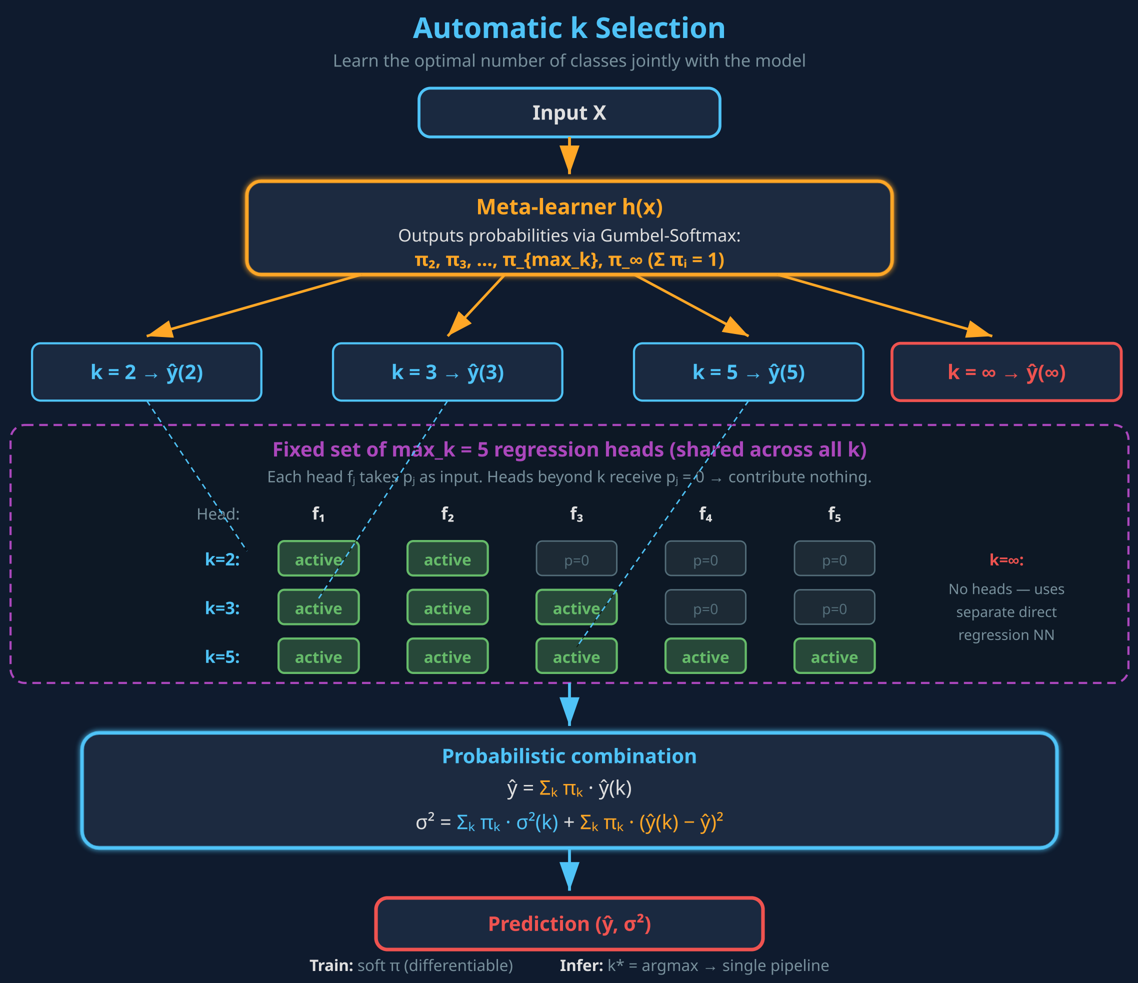
Learned Regression Functions



Each head $f_j(p_j)$ learns a distinct mapping from class probability to conditional mean

Automatic k Selection via Gumbel-Softmax

Can we **learn k from data** rather than treat it as a hyperparameter? — a differentiable architecture search problem.



Summary & Future Directions

Key Contributions

- Novel framework bridging classification & regression
- k as interpretable SNR proxy
- Calibrated uncertainty via Law of Total Variance
- Automatic k selection via Gumbel-Softmax

Relevance to Turing / AI for Physical Systems

- Calibrated uncertainty enables trustworthy surrogate models — scientists know when to run expensive simulations vs. trust the model
- Principled handling of noisy physical measurements
- Works with any classifier: GBMs, NNs, transformers

Future Work

- Theoretical analysis of optimal k ↔ SNR
- Benchmark vs Deep Ensembles, MC Dropout, CP
- Active learning: use σ^2 to guide data acquisition

Thank You

- Questions welcome